

# DAVID XCHEL MORALES HURTADO

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[www.linkedin.com/in/david-xchel-morales-hurtado](https://www.linkedin.com/in/david-xchel-morales-hurtado) | [dxchel.github.io](https://dxchel.github.io)

**Senior Software Reliability Developer** with 5+ years of professional experience building automation, validation and infrastructure tooling on **Linux** platforms. Strong background in **Python**, **C++**, automation and large scale test frameworks. Experienced in **developing** reliable engineering solutions, **mentoring** and **collaborating** across **global organizations**. Passionate about **open-source** software, **Mathematics** and continuous learning. Always aiming towards high **Quality** and **performance** with a **mentoring mindset** in favor of the continuous growth of the team.

## SELECTED ACHIEVEMENTS

- **Reduced validation execution times** from hours to minutes through **automation** using **Python** and **Bash**.
- **Improved testing efficiency** by 50% across processor validation programs.
- **Technical mentor** and **onboarding lead** for new engineers.
- **Bronze Medalist** – Mexican Mathematics Olympiad.
- **International academic exchange** at Universiti Malaya.

## PERSONAL PROJECTS

- Personal Arch Linux system, learning **Linux** and its **File System** deeply
- Personal **portfolio** found at [dxchel.github.io](https://dxchel.github.io), made using **React** and **TS**.
- **Continuously evolving C++ modular application** using **Gtk4**, **Threading**, **Coroutines** and **Network communication** with **GTEST testing**.

### Senior Software Reliability Developer

February 2026 — Present

Oracle, Guadalajara

Develop **automation** and **cloud**-management tooling in **Python**, Java and Bash for Oracle Cloud Infrastructure regions.

- Develop and maintain **automation** workflows aimed at reducing manual effort, improving **reliability** and **accelerating** cloud-region deployment activities.
- Troubleshooting of **Linux** systems and **cloud** infrastructure issues.
- **Developed** AI-assisted tooling for information extraction, documentation and operational support.
- **Create** technical workflows and onboarding material used across engineering teams.

### Functional Validation Engineer

February 2021 — February 2026

Intel, Guadalajara

Developed **Python** and **Bash** scripts made for the automated validation of next-generation processor architectures on **Linux** systems in collaboration with a **multicultural** and **diverse team** from all around the world.

- Helped **design** and **maintain** large-scale automation frameworks using **Python**, **Bash** and **PyTest**.
- **Reduced execution times from hours to minutes** through **automation** of manual workflows.
- **Created** migration tooling that reduced project-porting effort **from one day to minutes**.
- **Developed CI/CD**-based validation pipelines supporting multiple processor programs.
- Acted as **technical lead** and **point of contact** across teams.
- **Mentored** new engineers on processor architecture, memory systems and **Linux**-based development workflows.
- Improved validation **efficiency by 50%** through **optimization of test coverage** and execution strategies.

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## EDUCATION

### Mechatronics Engineer

August 2015 — January 2021

Instituto Politecnico Nacional, Mexico City

I was always a **top performing student** in this prestigious institution with **effort**, **curiosity** and **creativity**. Always focused on **teamwork** for the best outcome by **mediating** and **leading** delivering **quality** projects **efficiently**.

Relevant projects include:

- **Python** projects for **Neural Networks**, **file** and **database processing**, **web scraping**, Flask applications with Selenium testing.
- **MIPS** implementation in **System-Verilog** with **UVM testbench** creation using **python** and **C++**.
- **Virtual reality** experiences using **Unity** and **blender**.
- Embedded systems and FPGA development projects.
- Electronic circuits design and implementation.

Mobility program

Universiti Malaya, Kuala Lumpur

February 2019 — June 2019

Learned from one of the most renowned universities in the world while acquiring and adapting my experience working, **communicating** and **leading** people with contrasting backgrounds and mindsets.

Relevant projects include:

- Line follower with **Raspberry PI**.
- Hand tracking **AI** project.

SKILLS

Languages

**Python, C++, Bash**, C, Java, JavaScript, TypeScript, Verilog, Assembly

Platforms

**Linux**(Ubuntu, Debian, Arch), **Git**/**GitHub**, **Docker**

Engineering

**Agile**, **CI/CD**, **Automation**, **Testing Frameworks**, **Mathematics**, AI-assisted Development, Embedded Systems, Cloud

Professional

**Leadership**, **Mentoring**, **Technical Communication**, **Problem Solving**

Languages

**English**, Spanish, learning German

COURSES

|                                                        |                                   |
|--------------------------------------------------------|-----------------------------------|
| <b>Advanced Python, Universidad Marista</b>            | <b>June 2024 — November 2024</b>  |
| <b>Linux Essentials, Linux Professional Institute</b>  | <b>January 2020 — April 2020</b>  |
| <b>Advanced C++ Programming and Modern Practices</b>   | <b>February 2026 — May 2026</b>   |
| <b>Time Management Fundamentals</b>                    | <b>February 2026 — March 2026</b> |
| <b>Foundations of Agile Project Management</b>         | <b>February 2026 — March 2026</b> |
| <b>Learning how to Learn, Oakland University</b>       | <b>January 2025 — June 2025</b>   |
| Mobile App Development, Google Activate                | November 2019 — February 2020     |
| Intel x86 Processor & Platform Architecture, MindShare | January 2022 — June 2022          |

EXTRA-CURRICULAR ACTIVITIES

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|--------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Mexican Mathematics Olympiad, OMM, Mexico City</b><br>Bronze award.                                             | <b>March 2013 — November 2014</b> |
| <b>Mexican Informatics Olympiad, COMI, Mexico City</b><br>Participation award.                                     | <b>January 2014 — May 2014</b>    |
| Scout, Mexican Scout Association, Morelia<br>Participation in altruism and teamwork oriented activities            | January 2014 — August 2015        |
| Robot war staff, IEEE university branch, Mexico City<br>Staff in planning and organization of international event. | July 2018 — July 2020             |
| Emergency Response Helper, ERUM, Mexico<br>Provided help during the 2017 Mexico city earthquake                    | September 2017 — October 2017     |